TASK 8 – IMPLEMENT AND TEST THE PLANNING STRATEGIES (4 MARKS)

The following Python program is a reproducible reference simulator. It implements Erlang-B, valid cluster-size enumeration, hexagonal cell coordinates, first-tier Monte-Carlo SIR samples, and a simplified strategy comparison. It uses a fixed random seed so the same mobile positions and shadowing samples are used across strategies. The first-tier SIR model is intentionally stated as an approximation: a full engineering simulator should import actual antenna patterns, building clutter, measured path-loss maps and all co-channel sites.

IMPLEMENTATION:

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```
import math
import numpy as np

SEED = 2026
CHANNELS = 336
TARGET_B = 0.02

def erlang_b(m, A):
    b = 1.0
    for i in range(1, m + 1):
        b = A * b / (i + A * b)
    return b

def erlang_capacity(m, target=TARGET_B):
    lo, hi = 0.0, max(1.0, 3.0*m)
    for _ in range(100):
        mid = (lo + hi) / 2
        if erlang_b(m, mid) > target:
            hi = mid
        else:
            lo = mid
    return (lo + hi) / 2

def valid_clusters(limit=20):
    vals = {}
    for i in range(limit + 1):
        for j in range(limit + 1):
            n = i*i + i*j + j*j
            if 1 <= n <= limit:
                vals[n] = (i, j)
    return sorted(vals.items())

def hex_centres(rows, cols, R):
    # Pointy-top axial layout; nearest-neighbour spacing is sqrt(3)*R.
    pts = []
    for q in range(cols):
        for r in range(rows):
            x = math.sqrt(3)*R*(q + r/2)
            y = 1.5*R*r
            pts.append((x, y))
    return np.asarray(pts)

def first_tier_sir(N, n, samples=20000, sectors=1, seed=SEED):
    # Six idealised co-channel interferers at distance D=sqrt(3N)*R.
    # Random user positions are within a circular approximation of the serving cell.
    rng = np.random.default_rng(seed)
    R = 1.0
```

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    for a,b in zip(ix,iy):
        d = np.sqrt((x-a)**2+(y-b)**2)
        interf += d**(-n)
    # Ideal sector filtering: approximate removal of non-visible first-tier sites.
    # sectors=1 -> 6; sectors=3 -> 2; sectors=6 -> 1 dominant interferer.
    keep = {1:6, 3:2, 6:1}[sectors]
    # Retain the strongest 'keep' interferers per mobile.
    contributions = []
    for a,b in zip(ix,iy):
        d = np.sqrt((x-a)**2+(y-b)**2)
        contributions.append(d**(-n))
    strongest = np.sort(np.vstack(contributions), axis=0)[-keep:,:]
    interf = strongest.sum(axis=0)
    sir_db = 10*np.log10(desired/interf)
    return {
        "mean_dB": float(np.mean(sir_db)),
        "p5_dB": float(np.percentile(sir_db, 5)),
        "outage_pct_below_12dB": float(100*np.mean(sir_db < 12)),
        "outage_pct_below_18dB": float(100*np.mean(sir_db < 18))
    }

def compare(subscribers=(1000,10000,20000,100000)):
    rows = []
    for N in (3,4,7,12):
        m = CHANNELS // N
        cap = erlang_capacity(m)
        for users in subscribers:
            offered = users*0.025
            # Example 23 macro sites; after one uniform split 92 sites.
            for sites in (23,92):
                per_cell = offered/sites
                rows.append((N, users, sites, m, cap, per_cell,
                             erlang_b(m, per_cell)))
    return rows

if __name__ == "__main__":
    print("Valid clusters:", valid_clusters())
    for n in (2.5,3,4):
        for N in (3,4,7,12):
            for s in (1,3,6):
                print("SIR", N, n, "sectors", s,
                      first_tier_sir(N,n,sectors=s))
    for row in compare():
        print("N,users,sites,m,capacity,load/cell,blocking:", row)
    print("Sanity N=7,n=4:",
          10*math.log10((3*7)**(4/2)/6), "dB")
```

OUTPUT:

```
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Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug  6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/monisha D/AppData/Local/Programs/Python/Python312/MC-ASS.py
Valid clusters: [(1, (1, 0)), (3, (1, 1)), (4, (2, 0)), (7, (2, 1)), (9, (3, 0)), (12, (2, 2)), (13, (3, 1)), (16, (4, 0)), (19, (3, 2))]
SIR 3 2.5 sectors 1 {'mean_dB': 9.113370654097505, 'p5_dB': 3.661585823211058, 'outage_pct_below_12dB': 77.735, 'outage_pct_below_18dB': 92.75999999999999}
SIR 3 2.5 sectors 3 {'mean_dB': 12.114050630873747, 'p5_dB': 5.96034147351432, 'outage_pct_below_12dB': 60.785, 'outage_pct_below_18dB': 85.395}
SIR 3 2.5 sectors 6 {'mean_dB': 14.687319854170902, 'p5_dB': 8.233264001818082, 'outage_pct_below_12dB': 42.575, 'outage_pct_below_18dB': 77.005}
SIR 4 2.5 sectors 1 {'mean_dB': 10.776017581421288, 'p5_dB': 5.420880481668408, 'outage_pct_below_12dB': 69.77499999999999, 'outage_pct_below_18dB': 90.17500000000001}
SIR 4 2.5 sectors 3 {'mean_dB': 13.98751221414788, 'p5_dB': 8.00490993592481, 'outage_pct_below_12dB': 47.85, 'outage_pct_below_18dB': 80.435}
SIR 4 2.5 sectors 6 {'mean_dB': 16.63545237618456, 'p5_dB': 10.413940407348488, 'outage_pct_below_12dB': 23.325000000000003, 'outage_pct_below_18dB': 68.83500000000001}
SIR 7 2.5 sectors 1 {'mean_dB': 13.940900566959124, 'p5_dB': 8.705558400110212, 'outage_pct_below_12dB': 47.025, 'outage_pct_below_18dB': 82.505}
SIR 7 2.5 sectors 3 {'mean_dB': 17.501616433810977, 'p5_dB': 11.76206586471073, 'outage_pct_below_12dB': 8.725, 'outage_pct_below_18dB': 64.905}
SIR 7 2.5 sectors 6 {'mean_dB': 20.25679712262995, 'p5_dB': 14.353579682831654, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 45.385}
SIR 12 2.5 sectors 1 {'mean_dB': 16.93612668408575, 'p5_dB': 11.764449318044248, 'outage_pct_below_12dB': 9.120000000000001, 'outage_pct_below_18dB': 69.265}
SIR 12 2.5 sectors 3 {'mean_dB': 20.765186885465006, 'p5_dB': 15.193286885518326, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 39.86}
SIR 12 2.5 sectors 6 {'mean_dB': 23.590755627314344, 'p5_dB': 17.90171810888234, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 6.375}
SIR 3 3 sectors 1 {'mean_dB': 12.400149961303484, 'p5_dB': 5.772554208609795, 'outage_pct_below_12dB': 59.714999999999996, 'outage_pct_below_18dB': 83.705}
SIR 3 3 sectors 3 {'mean_dB': 15.129390056060952, 'p5_dB': 7.732617186847573, 'outage_pct_below_12dB': 42.975, 'outage_pct_below_18dB': 73.935}
SIR 3 3 sectors 6 {'mean_dB': 17.624783825500508, 'p5_dB': 9.879916802181699, 'outage_pct_below_12dB': 24.195, 'outage_pct_below_18dB': 62.875}
SIR 4 3 sectors 1 {'mean_dB': 14.418671290590447, 'p5_dB': 7.929250019885447, 'outage_pct_below_12dB': 46.394999999999996, 'outage_pct_below_18dB': 78.175}
SIR 4 3 sectors 3 {'mean_dB': 17.380725702729606, 'p5_dB': 10.190453662213399, 'outage_pct_below_12dB': 23.549999999999997, 'outage_pct_below_18dB': 64.625}
SIR 4 3 sectors 6 {'mean_dB': 19.96254285142147, 'p5_dB': 12.496728488818183, 'outage_pct_below_12dB': 0.91, 'outage_pct_below_18dB': 50.54}
SIR 7 3 sectors 1 {'mean_dB': 18.24624722598983, 'p5_dB': 11.928173570511008, 'outage_pct_below_12dB': 6.085, 'outage_pct_below_18dB': 60.985}
SIR 7 3 sectors 3 {'mean_dB': 21.600996008464787, 'p5_dB': 14.710101237227434, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 38.155}
SIR 7 3 sectors 6 {'mean_dB': 24.308156547155942, 'p5_dB': 17.224295619397985, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 13.525}
SIR 12 3 sectors 1 {'mean_dB': 21.856889923555595, 'p5_dB': 15.630335482386808, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 32.985}
SIR 12 3 sectors 3 {'mean_dB': 25.518758141669807, 'p5_dB': 18.83150646071102, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 3 sectors 6 {'mean_dB': 28.308906752777215, 'p5_dB': 21.482061730658803, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 3 4 sectors 1 {'mean_dB': 18.89040686447007, 'p5_dB': 9.839730269041885, 'outage_pct_below_12dB': 22.509999999999998, 'outage_pct_below_18dB': 57.835}
SIR 3 4 sectors 3 {'mean_dB': 21.15097825315193, 'p5_dB': 11.250298924731828, 'outage_pct_below_12dB': 11.105, 'outage_pct_below_18dB': 48.1}
SIR 3 4 sectors 6 {'mean_dB': 23.499711766673443, 'p5_dB': 13.173222402908893, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 36.265}
SIR 4 4 sectors 1 {'mean_dB': 21.640168516734754, 'p5_dB': 12.828502194296751, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 44.015}
SIR 4 4 sectors 3 {'mean_dB': 24.160998907676245, 'p5_dB': 14.551763546797979, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 30.605}
SIR 4 4 sectors 6 {'mean_dB': 26.61672380189529, 'p5_dB': 16.66230465175758, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 14.985000000000001}
SIR 7 4 sectors 1 {'mean_dB': 26.819470256469078, 'p5_dB': 18.302545496425736, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 1.915}
SIR 7 4 sectors 3 {'mean_dB': 29.79679481842565, 'p5_dB': 20.599218058624842, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 7 4 sectors 6 {'mean_dB': 32.41087539620792, 'p5_dB': 22.965727492530647, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 4 sectors 1 {'mean_dB': 31.67622266064375, 'p5_dB': 23.320130361781874, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 4 sectors 3 {'mean_dB': 35.02438563340119, 'p5_dB': 26.10102234299166, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 4 sectors 6 {'mean_dB': 37.74520900370295, 'p5_dB': 28.64274897421174, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
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N,users,sites,m,capacity,load/cell,blocking: (3, 10000, 92, 112, 99.62382110612876, 2.7173913043478261, 1.4107095734438536e-135)
N,users,sites,m,capacity,load/cell,blocking: (3, 20000, 23, 112, 99.62382110612876, 21.739130434782608, 1.0826293986863974e-42)
N,users,sites,m,capacity,load/cell,blocking: (3, 20000, 92, 112, 99.62382110612876, 5.434782608695652, 4.837813023386688e-103)
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IDLE Shell 3.12.5

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SIR 12 3 sectors 6 {'mean_dB': 28.308906752777215, 'p5_dB': 21.482061730658803, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 3 4 sectors 1 {'mean_dB': 18.89040686447007, 'p5_dB': 9.839730269041885, 'outage_pct_below_12dB': 22.509999999999998, 'outage_pct_below_18dB': 57.835}
SIR 3 4 sectors 3 {'mean_dB': 21.15097825315193, 'p5_dB': 11.250298924731828, 'outage_pct_below_12dB': 11.105, 'outage_pct_below_18dB': 48.1}
SIR 3 4 sectors 6 {'mean_dB': 23.499711766673443, 'p5_dB': 13.173222402908893, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 36.265}
SIR 4 4 sectors 1 {'mean_dB': 21.640168516734754, 'p5_dB': 12.828502194296751, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 44.015}
SIR 4 4 sectors 3 {'mean_dB': 24.160998907676245, 'p5_dB': 14.551763546797979, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 30.605}
SIR 4 4 sectors 6 {'mean_dB': 26.61672380189529, 'p5_dB': 16.66230465175758, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 14.985000000000001}
SIR 7 4 sectors 1 {'mean_dB': 26.819470256469078, 'p5_dB': 18.302545496425736, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 1.915}
SIR 7 4 sectors 3 {'mean_dB': 29.79679481842565, 'p5_dB': 20.599218058624842, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 7 4 sectors 6 {'mean_dB': 32.41087539620792, 'p5_dB': 22.965727492530647, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 4 sectors 1 {'mean_dB': 31.67622266064375, 'p5_dB': 23.320130361781874, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 4 sectors 3 {'mean_dB': 35.02438563340119, 'p5_dB': 26.10102234299166, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
SIR 12 4 sectors 6 {'mean_dB': 37.74520900370295, 'p5_dB': 28.64274897421174, 'outage_pct_below_12dB': 0.0, 'outage_pct_below_18dB': 0.0}
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N,users,sites,m,capacity,load/cell,blocking: (7, 100000, 92, 48, 38.39164117230078, 27.173913043478261, 8.789429463380738e-05)
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N,users,sites,m,capacity,load/cell,blocking: (12, 100000, 23, 28, 20.150377704053582, 108.69565217391305, 0.7454914493631855)
N,users,sites,m,capacity,load/cell,blocking: (12, 100000, 92, 28, 20.150377704053582, 27.173913043478261, 0.12133259422949884)
Sanity N=7,n=4: 18.66287339084195 dB
```